

CUSTOMER ORDER AND PURCHASE ANALYSIS SYSTEM

1. Introduction

The Customer Order and Purchase Analysis System is used to store and analyse customer orders.

The system keeps information about:

- Customers
- Products
- Orders
- Products included in each order

Using SQL queries, we can find the order history of customers, product quantities, and customer spending.

2. Tables Used

Four tables are used in this system:

CUSTOMER

The CUSTOMER table stores customer information.

It contains:

- Customer ID
- First Name
- Last Name

```
SQL> CREATE TABLE CUSTOMER (Customer_ID VARCHAR2(10) PRIMARY KEY, First_Name VARCHAR2(50), Last_Name VARCHAR2(50));  
Table created.
```

PRODUCT

The PRODUCT table stores product information.

It contains:

- Product ID
- Product Name
- Price
- Stock

```
SQL> CREATE TABLE PRODUCT (Product_ID NUMBER PRIMARY KEY, Product_Name VARCHAR2(50) NOT NULL, Price NUMBER(10,2), Stock NUMBER);  
Table created.
```

ORDERS

The ORDERS table stores customer order information.

It contains:

- Order ID
- Customer ID
- Order Date
- Total Amount
- Order Status

```
SQL> CREATE TABLE ORDERS (Order_ID VARCHAR2(10) PRIMARY KEY, Customer_ID VARCHAR2(10), Order_Date DATE NOT NULL, Total_Amount NUMBER(10,2), Order_Status VARCHAR2(20));  
Table created.
```

ORDER_DETAILS

The ORDER_DETAILS table stores the products included in each order.

It contains:

- Order Detail ID
- Order ID
- Product ID
- Quantity
- Price

```
SQL> CREATE TABLE ORDER_DETAILS (Order_Detail_ID VARCHAR2(10) PRIMARY KEY, Order_ID VARCHAR2(10), Product_ID NUMBER, Quantity NUMBER, Price NUMBER(10,2));  
Table created.
```

3. Constraints Used

Constraints are used to keep the data correct.

The following constraints were used:

- **Primary Key** – uniquely identifies each record.
- **Foreign Key** – connects two related tables.
- **NOT NULL** – makes sure a required value is entered.
- **CHECK** – makes sure the entered value follows a condition.

For example, CUSTOMER_ID is used to identify each customer, and ORDER_ID identifies each order.

4. Customer Order History Report

The Customer Order History Report shows the order details of each customer.

It displays:

- Customer Name
- Order ID
- Order Date
- Total Amount
- Order Status

This is one of the reports required in the Week 4 document.

Query

```
SQL> SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,O.ORDER_ID,O.ORDER_DATE,O.TOTAL_AMOUNT,O.ORDER_STATUS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID;
```

Result

CUSTOMER_NAME	ORDER_ID	ORDER_DATE	TOTAL_AMOUNT	ORDER_STATUS
Aarav Kumar	0101	28-SEP-26	1500.00	Pending
Diya Sharma	0102	28-SEP-26	2500.00	Shipped
sanjay prakash	0103	28-SEP-26	1200.00	Delivered
Moni Sha	0104	28-SEP-26	1800.00	Pending

5. Product-wise Order Report

The Product-wise Order Report shows how many times each product was ordered and the total quantity sold.

5.1 Number of Times Each Product Was Ordered

Query

```
SQL> SELECT P.PRODUCT_NAME,COUNT(O.ORDER_DETAIL_ID) AS TIMES_ORDERED FROM PRODUCT P JOIN ORDER_DETAILS O ON P.PRODUCT_ID=O.PRODUCT_ID GROUP BY P.PRODUCT_NAME;
```

Result

PRODUCT_NAME	TIMES_ORDERED
Organic Apples	1
Brown Rice	1
Carrots	1
Bananas	1

5.2 Total Quantity of Each Product Sold

Query

```
SQL> SELECT P.PRODUCT_NAME,SUM(O.QUANTITY) AS TOTAL_QUANTITY FROM PRODUCT P JOIN ORDER_DETAILS O ON P.PRODUCT_ID=O.PRODUCT_ID GROUP BY P.PRODUCT_NAME;
```

Result

PRODUCT_NAME	TOTAL_QUANTITY
Organic Apples	5
Brown Rice	10
Carrots	8
Bananas	6

6. Customer Purchase Analysis

Customer Purchase Analysis is used to understand the purchasing activity of customers.

6.1 Customer with Maximum Orders

This query finds the customer who has placed the maximum number of orders.

Query

```
SQL> SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME, COUNT(O.ORDER_ID) AS TOTAL_ORDERS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID GROUP BY C.FIRST_NAME, C.LAST_NAME ORDER BY TOTAL_ORDERS DESC FETCH FIRST 1 ROW ONLY;
```

Result

CUSTOMER_NAME	TOTAL_ORDERS
Aarav Kumar	1

In the current data, every customer has one order. The query displays Aarav Kumar because it returns the first row when there is a tie.

6.2 Total Spending by Each Customer

This report shows the total amount spent by each customer.

Query

```
SQL> SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME, SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID GROUP BY C.FIRST_NAME, C.LAST_NAME;
```

Result

CUSTOMER_NAME	TOTAL_SPENDING
Aarav Kumar	1500
Diya Sharma	2500
sanjay prakash	1200
Moni Sha	1800

6.3 Average Order Value

This report shows the average value of orders placed by each customer.

Query

```
SQL> SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME, AVG(O.TOTAL_AMOUNT) AS AVERAGE_ORDER_VALUE FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID GROUP BY C.FIRST_NAME, C.LAST_NAME;
```

Result

CUSTOMER_NAME	AVERAGE_ORDER_VALUE
Aarav Kumar	1500
Diya Sharma	2500
sanjay prakash	1200
Moni Sha	1800

Because each customer currently has only one order, the average order value is the same as that customer's order amount.

7. Conclusion

The Customer Order and Purchase Analysis System was successfully created using SQL.

Four tables were created: CUSTOMER, PRODUCT, ORDERS, and ORDER_DETAILS.

The required reports were generated successfully:

- Customer Order History
- Product-wise Order Report
- Customer with Maximum Orders
- Total Spending by Each Customer
- Average Order Value

These reports help us understand customer orders, product quantities, and customer spending.